# Alicia Martín

DPhil Astrophysics, University of Oxford
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## Summary

I develop interpretable machine learning methods for scientific discovery. My main focus is symbolic regression, which finds closed-form models directly from data. I apply these methods to dark matter halo structure using gravitational lensing and galaxy dynamics, and to the design of informative summary statistics for large-scale structure. More broadly, I am interested in scalable and statistically principled methods for model discovery, optimisation and Bayesian inference, using challenging problems in astrophysics and cosmology to drive advances in machine learning.

## Education

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|---|---|---|
| 2023 – present (expected 2027) | DPhil (PhD) Astrophysics at **University of Oxford**<br>*Thesis:* Symbolic regression, machine learning and cosmology: deriving interpretable, closed-form expressions for physical quantities.<br>`Supervisors`: Pedro G. Ferreira, Harry Desmond, Deaglan J. Bartlett | |
| 2019 – 2023 | MPhys (Hons) Physics, First Class, at **University of Manchester** | Top 10% of class |

## Honours

## Publications

*First author on all papers.*

**Published**

1. **Martín, A.**, Yasin, T., Bartlett, D. J., Desmond, H. & Ferreira, P. G. Symbolically regressing dark matter halo profiles using weak lensing. *MNRAS* **551**, stag1394 (2026). arXiv: 2601.05203. DOI: 10.1093/mnras/stag1394.
2. **Martín, A.**, Yasin, T., Bartlett, D. J., Desmond, H. & Ferreira, P. G. Constraining dark matter halo profiles with symbolic regression. *Phil. Trans. R. Soc. A* **384**, 20250090 (2026). arXiv: 2511.23073. DOI: 10.1098/rsta.2025.0090.
3. **Martín, A.**, Skordis, C., Bartlett, D. J., Desmond, H., Ferreira, P. G. & Yasin, T. Cosmological dipole in tilted anisotropic universes. *Phys. Rev. D* **114**, 023526 (2026). arXiv: 2512.03867. DOI: 10.1103/82sh-h3cv.

**In preparation**

4. **Martín, A.** et al. Symbolically regressing dark matter halo profiles of CLASH galaxy clusters. In preparation.
5. **Martín, A.** & Varasteanu, A. Symbolically regressing dark matter halo profiles from MIGHTEE rotation curves. In preparation.
6. **Martín, A.** et al. Optimal mark functions for weak-lensing marked power spectra. In preparation.

## Selected Talks and Presentations

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|---|---|
| *Symbolically regressing dark matter halo profiles using weak lensing* — ML4Astro Conference, Malta | 31 Aug – 4 Sep 2026 |
| *Symbolically regressing dark matter halo profiles using weak lensing* — AstroAI Workshop 2026, Center for Astrophysics \| Harvard & Smithsonian | Jun 2026 |
| *Symbolic regression: inferring dark matter halo profiles from weak lensing* — UK Cosmology Meeting, Imperial College London | Mar 2026 |
| *Empirical dark matter halo profiles from weak lensing* — RAS 3rd Machine Learning and AI applied to Astronomy meeting, Burlington House, London | Jan 2026 |
| *Empirical dark matter halo profiles* — National Astronomy Meeting (NAM 2025), Durham | Jul 2025 |
| *Learning dark matter halo profiles using symbolic regression* — National Astronomy Meeting (NAM 2024), Hull | Jul 2024 |

## Prior Research Experience

**Modified gravity, Master's thesis — Manchester (UK)** Sep 2022 – Jun 2023
Fifth forces and symmetron screening in scalar-tensor gravity. `Supervisor`: Dr Peter Millington

**ML and dark matter, Student Researcher — Oxford (UK)** Jul 2022 – Sep 2023
Bayesian/MCMC inference of dark matter profiles from SPARC data; precursor to my current DPhil project. `Supervisors`: Prof Pedro Ferreira, Dr Harry Desmond

**Quantum gravity, Student Researcher — Penn State (USA, remote)** Aug 2021 – Oct 2022
Quantum corrections to the classical Hamiltonian in loop quantum gravity. `Supervisor`: Prof Martin Bojowald

**Black hole dynamics, Student Researcher — Cambridge (UK)** Jun 2021 – Feb 2022
Thermal conduction and magnetic fields in supermassive black hole dynamics. `Supervisors`: Dr Ricarda Beckmann, Dr Debora Sijacki

## Teaching Experience

**College Tutor, Wadham College, University of Oxford** 2024 – 2026
Symmetry and Relativity (3rd-year course): weekly tutorials for two years, marking problem sets and test exams.

**Laboratory Demonstrator, University of Oxford** 2023 – 2025
Supervised 3rd-year students in the astrophysics teaching labs for two years.

## Other Experience

**Graduate Student Representative — Astrophysics, Oxford** 2024 – 2025
Represented DPhil students' interests to departmental staff and committees for one academic year.

**START school in theoretical physics — Perimeter Institute (Canada)** May – Aug 2022
Ten-week online school in theoretical physics: quantum information, path integrals, symmetries and numerical methods. Final project on quantum foundations and the Spekkens toy model.

## Awards

2023 **Stellify Award**, The University of Manchester. Award for leadership and volunteering, recognising my work as President of the Physics Society and community volunteering.

2022 **Nova 111 under 25 Award**, Nova Talent. Selected as one of the 111 highest-potential students under 25 in Spain, 10 per discipline (Mathematics, Physics & Chemistry).

2019 **Excellence Scholarship**, Spanish government. Awarded to the top 1% of undergraduate students.

## Skills

**Machine learning**: symbolic regression (exhaustive: ESR; genetic programming: Operon), Gaussian processes, gradient-based optimisation, neural networks, normalising flows.

**Statistics & inference**: Bayesian inference, MCMC, HMC model selection (MDL), optimisation.

**Programming**: Python, NumPy, SciPy, JAX, Mathematica, Git/GitHub, LaTeX.